

AROGUNDADE MUIDEEN TUNDE

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EDUCATION

Master of Science – Biochemistry

Awarded September, 2024

Osun State University, Osogbo, Nigeria.

Score: **66.32%**

Master Degree Thesis: **Therapeutic potential of *Ocimumgratissimum* biogenic zinc oxide nanoparticles on rotenone-exposed *Drosophila melanogaster* Parkinson's disease model.**

Supervisor: Dr. O. O. Adewale & Dr. G. T. Atere

Bachelor of Science – Biochemistry

Awarded March, 2018

Osun State University, Osogbo, Nigeria.

CGPA: **3.88/5**

Undergraduate Thesis: **Modulatory effect of *Buchholziacoriaceamethanolic* extract on aluminum chloride-induced hepatotoxicity in male wistar rat.**

Supervisor: Dr. B. F. Olabiyi

RESEARCH INTEREST

- Investigating the Molecular Mechanisms of Neurodegeneration and Develop Approaches to Prevent or Treat Them

RESEARCH EXPERIENCE

Research Assistant

June 2024 – Present

Neuro-therapeutics Lab, Department of Medical Biochemistry, Osun State University, Osogbo Nigeria

- Mentoring undergraduate students on research methodology
- Set up an empirical research on the beneficial effect of natural products on aging and neurodegenerative diseases using mice and fruit fly as a research model
- Analyzing research data using Graph pad prism and preparing the manuscript

Graduate Research Assistant

October 2023 – May 2024

Drosophila Lab, Department of Medical Biochemistry, Osun State University, Osogbo Nigeria

- Bio-synthesized nanoscale particles of different metals like Zinc oxide, Silver e.t.c.

- Physically characterized nanoparticles using UV-VIS Spec., SEM, TEM and XRD techniques.
- Designed and performed an empirical research on the ameliorative role of green synthesized metallic nano-particles on neurodegenerative disorders using fruit fly as a model
- Performed behavioral assays, oxidative stress assays, antioxidant screening and some other biochemical assays using Light Microscope, Centrifuge, High performance liquid chromatography and Spectrometer, as the major analytical instruments.

Research Fellow

17th- 30th August 2023

BioRTC-ISN Pan Africa Summer School,

Yobe State University, Damaturu Nigeria.

Supervisor: Professor Louise Serpell, University of Sussex, UK.

- Performed cell culturing techniques such as, media preparation, cell line maintenance, cell passage, cell counting, sub-culturing, aseptic technique, e.t.c.
- Protein visualization and localization with immunocytochemistry and western blotting techniques with confocal and fluorescence microscopy.
- Trained on Open Source Hardware such as, 3D printing, Open UC2 microscopy and Fiji software.

Undergraduate Research Assistant

September 2016 – August 2017

Toxicology Laboratory, Osun State University

Supervisor: Dr (Mrs) Fatimat B. Olabiyi

- Extracted and performed phytochemical analysis on plants, such as *Buchholziacoriaceae*
- Handled, administered and sacrificed wistar rats as a research model and Performed biochemical assays on liver homogenate using Centrifuge, High performance liquid chromatography and Spectrometer, as the major analytical instruments.

TEACHING EXPERIENCE

Teaching Assistant

30th Oct. 2024 – 6th Mar. 2025

Neurobiology SF2402 Summer Course at SCIENSPUR, India.

Chemistry and Biology Tutor

Sep. 2021– Oct. 2022

Ife-oluwa Senior Secondary School, Osogbo, Osun State. Under Federal Government-Npower Scheme.

Chemistry Tutor

Jul. 2018- Jun. 2019

Obokun Senior Secondary School, Ibokun, Osun State. National Youth Service Corps (NYSC).

SELECTED AWARDS

- DELSU Joint Canada-Israel Neuroscience & Biopsychiatry Laboratory NSN (Neuroscience Society of Nigeria) 2024 Conference Support Grant **June 2024**
- Best Poster Award at Neuroscience Diversity Promotion in Pakistan. **December 2023**
- BioRTC-ISN 2023 Summer School Fellowship **August 2023**

CONFERENCES

- Neuroscience Society of Nigeria (NSN) 21st Annual Conference (Oral and Poster presented) **3rd – 6th September 2024**
- Neuroscience Diversity Promotion in Pakistan (Poster presented) **December 2023**
- 7th Annual Meeting of the International Society for Neurodegenerative Disease. **8th – 9th September 2023**

TRAINING AND PROFESSIONAL DEVELOPMENT

- Flycross Mentorship Program (Under the mentorship of Dr. Elizabeth Brown of Florida University, Florida, USA). **Feb. 2025 – Present**
- SUU Data Science Workshop: Python Programming for Beginners **Mar. 2025 – Apr. 2025**
- Tableau Master Class for Data Analysis and Visualization at Ultra-Ukay Tech Systems **05th May – 3rd June 2024**
- Parkinson's Genomics at Gen'Omics Research Hub **5th July 2024- 24th August 2023**
- Introduction to Bioinformatics at H3ABioNet, South Africa **2nd April– 25th July 2024**
- SCIENSPUR India 2024 Winters School: Developmental Biology Course **3rd March – 30th April 2024**
- SCIENSPUR India 2023 Fall School: Neurobiology. **14th October 2023- 30th January 2024**
- MDS-AS 3rd Virtual Interactive Course: Principle of Movement Disorder by International Parkinson's Movement Disorder Society. **20th June – 30th August 2023**
- SCIENSPUR India 2023 Summer School: New Age Molecular Genetics **2nd June- 30th July 2023**
- Africa Young Scientists Mentorship Program (AYSMP) **November 2022- October 2023**

PUBLICATION

Journals

- ATERE, T. G., WAHYUNI, D. K., AVWIOROKO, O. J., BAYO-OLUGBAMI, A. A., IDOWU, O. K., AROGUNDADE, M. T., ... & ADEREMI, A. V. (2025). Proanthocyanidin from grape seed against rotenone-induced toxicity in *Drosophila melanogaster*: Insights from behavioural and antioxidant studies. *NotulaeScientiaBiologicae*, 17(1), 12341-12341. <https://doi.org/10.55779/nsb17112341>
- Therapeutic potential of *Ocimumgratissimum* biogenic zno-nanoparticles on rotenone-induced neurotoxicity in *Drosophila melanogaster*. Submitted to BioNanoScience Journal (Submission ID e72c352a-1ecc-4c23-8543-16295dbae0f1). **Accepted for Publication.**
- Daucuscarota L seed oil possesses anti-inflammatory properties, mitigates ageing-related oxidative stress and prolong lifespan in D-galactose-treated *Drosophila melanogaster*. Submitted to Chemistry and Biodiversity Journal (Submission ID cbdv.202500916). **Under Review**

Science Communication

- How to Choose the Right Model for Alzheimer's Disease

<https://ghscientific.com/right-model-for-alzheimers-disease-study/>

PROFESSIONAL MEMBERSHIPS

- **Member:** Neuroscience Society of Nigeria (NSN)
- **Member:**International Youth Neuroscience Association (IYNA)
- **Member:** Africa Young Scientist (AYS)
- **Member:**The Nigeria Society of Biochemistry and Molecular Biology
- **Member:**Lautech Neuroscience Group (LNG)

REFEREES

(1) Dr Bolanle Fatimot Olabiyi

bfo21@cam.ac.uk

Relationship: B.Sc. Project Supervisor

(2) Dr Tope Gafar Atere

tope.ater@uniosun.edu.ng

Relationship: M.Sc. Project Supervisor